# DEPTH INEQUALITIES ALONG A SHORT EXACT SEQUENCE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The characterization of depth through the vanishing of Ext groups against the quotient by the ideal. Invoked inside the counted proof (source lines 2014-2017). Replaces each depth by the first index where an Ext group is nonzero.
- The long exact cohomology sequence for Ext groups. Invoked inside the counted proof (source lines 2018-2028) as a cross-chapter pointer. Relates the three families of Ext groups.

*Source reference.* Original reference algebra-lemma-long-exact-seq-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 2032–2032. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-long-exact-seq-ext, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

*Source reference.* Original lemma lemma-depth: sources/stacks-dualizing/pinned-original-source.tex, source lines 1921–1921. Navigation pointer only: the counted text refers to the Stacks item lemma-depth, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

**Lemma 0.1.** *Let $A$ be a Noetherian ring. Let $0 \to N' \to N \to N'' \to 0$ be a short exact sequence of finite $A$-modules. Let $I \subset A$ be an ideal.*

(1) $\mathit{depth}_I(N) \geq \min\{\mathit{depth}_I(N'), \mathit{depth}_I(N'')\}$
(2) $\mathit{depth}_I(N'') \geq \min\{\mathit{depth}_I(N), \mathit{depth}_I(N') - 1\}$

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(3) $depth_I(N') \geq \min\{depth_I(N), depth_I(N'') + 1\}$

*Proof.* Assume $IN \neq N$, $IN' \neq N'$, and $IN'' \neq N''$. Then we can use the characterization of depth using the Ext groups $\mathrm{Ext}^i(A/I, N)$, see Lemma lemma-depth, and use the long exact cohomology sequence

$$\begin{aligned} 0 \to \mathrm{Hom}_A(A/I, N') \to \mathrm{Hom}_A(A/I, N) \to \mathrm{Hom}_A(A/I, N'') \\ \to \mathrm{Ext}^1_A(A/I, N') \to \mathrm{Ext}^1_A(A/I, N) \to \mathrm{Ext}^1_A(A/I, N'') \to \ldots \end{aligned}$$

from Algebra, Lemma algebra-lemma-long-exact-seq-ext. This argument also works if $IN = N$ because in this case $\mathrm{Ext}^i_A(A/I, N) = 0$ for all $i$. Similarly in case $IN' \neq N'$ and/or $IN'' \neq N''$. □